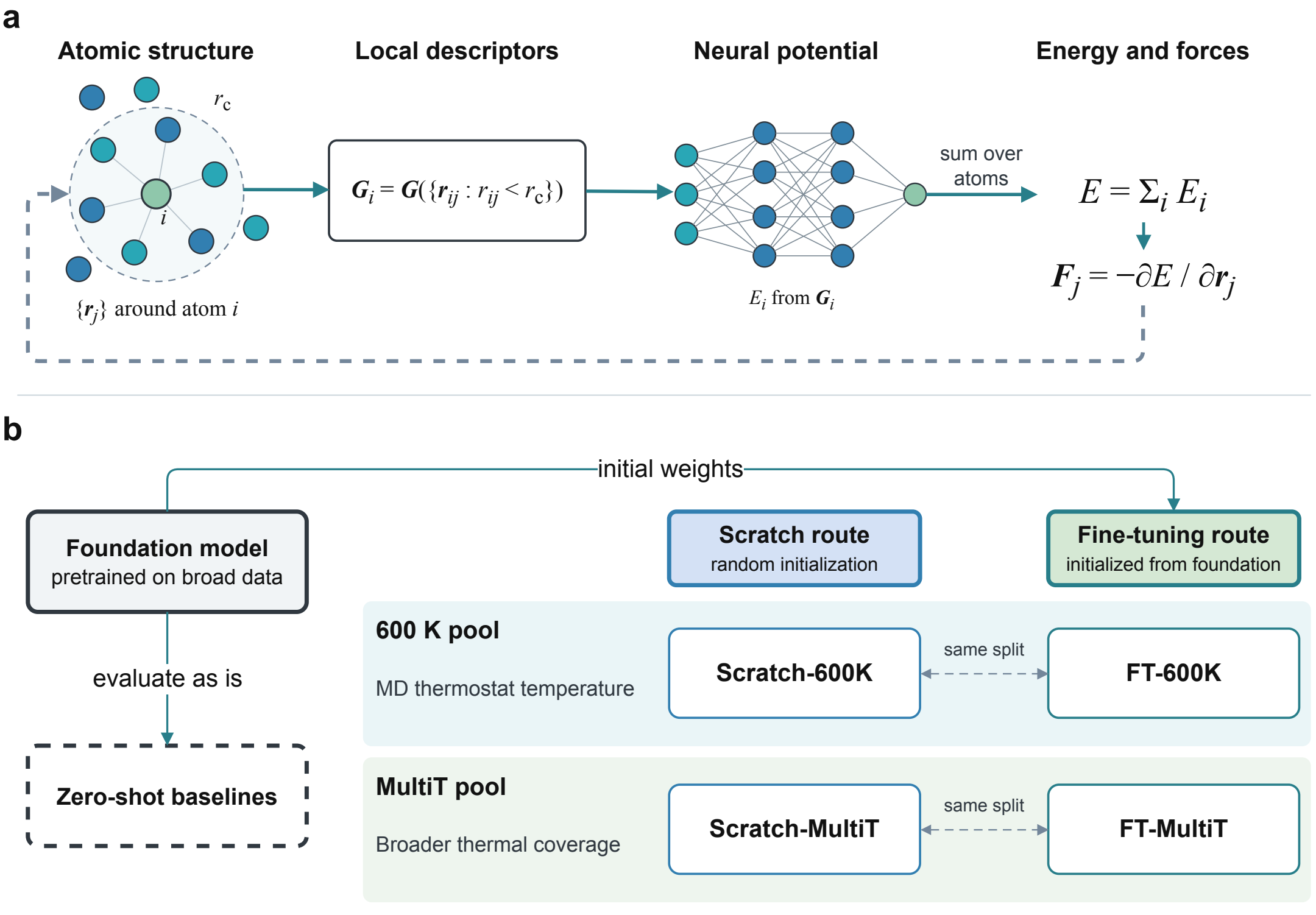

a
Atomic structure
Local descriptors
Neural potential
Energy and forces
r_c
i
$\{r_j\}$ around atom i
$G_i = G(\{r_{ij} : r_{ij} < r_c\})$
E_i from G_i
sum over atoms
$E = \Sigma_i E_i$
$F_j = -\partial E / \partial r_j$
b
initial weights
Foundation model
pretrained on broad data
evaluate as is
Zero-shot baselines
Scratch route
random initialization
Fine-tuning route
initialized from foundation
600 K pool
MD thermostat temperature
Scratch-600K
same split
FT-600K
MultiT pool
Broader thermal coverage
Scratch-MultiT
same split
FT-MultiT